

Parental Perceptions of Children's d/Deaf Identity Shaping Technology Use: A Qualitative Study on Communication Technologies in Mixed-hearing Families

Keyi Zeng
Thrust of Computational Media and Arts
The Hong Kong University of Science and Technology (Guangzhou)
Guangzhou, China
keyizeng@hkust-gz.edu.cn

Jingyang Lin
Duke Kunshan University
Kunshan, Jiangsu, China
jl1198@duke.edu

Ruiqi Chen
Human Centered Design & Engineering
University of Washington
Seattle, Washington, USA
ruiqich@uw.edu

RAY LC
School of Creative Media
City University of Hong Kong
Hong Kong, China
lc@raylc.org

Pan Hui
Computational Media and Arts
The Hong Kong University of Science and Technology
Hong Kong, China
panhui@ust.hk

Xin Tong*
Thrust of Computational Media and Arts
Hong Kong University of Science and Technology (Guangzhou)
Guangzhou, China
xint@hkust-gz.edu.cn

Abstract

Various assistive technologies have been developed to facilitate communication between deaf and hard-of-hearing (DHH) individuals and hearing people. However, as mixed-hearing families (i.e., hearing parents with DHH children or DHH parents with hearing children) encounter unique communication dynamics, the limited studies explored the use and perception of technologies in such families. To explore this question, we conducted semi-structured interviews with eleven parents from mixed-hearing families, examining their current use of technology in communication, as well as their perceptions, challenges, and expectations. Our findings show that spoken language is the primary communication method, with parents prioritizing spoken language education over sign language. Additionally, current technology use, expectations, and potential consequences were identified. This study contributes empirical insights and design implications to enhance communication technologies for mixed-hearing families.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Parent-Child communication, mixed-hearing family, assistive technology, augmentative alternative communication

*Corresponding authors

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1 INTRODUCTION

Similar to D/deaf¹ and hard-of-hearing (DHH) and hearing individuals, mixed-families face unique communication dynamics [46, 63, 70]. Mixed-families refer to families where individuals have different hearing statuses (such as hearing parents with DHH children or DHH parents with hearing children). Parents and children in these families have differences in auditory perception and languages, leading to unique communication dynamics and potential communication difficulties [28, 47, 60, 81]. Communication in families goes beyond basic messaging and affects the well-being, self-identity, and development of family members [10, 37, 48, 69], and communication difficulties can increase emotional and behavioral problems [22, 47, 67].

Recent studies have investigated the use of technologies such as captions [35, 45, 55], real-time interpretation [8, 41], and augmented reality [52, 53] to facilitate communication between DHH and hearing individuals [54]. However, limited research has explored the application of these technologies in mixed-hearing families. As parents need to communicate with children in early language development in families, they encounter different communication dynamics from adults' communication. Understanding parents' perceptions of technologies will help future research to facilitate communication in mixed-hearing families. However, limited studies explored facilitating sign language communication through technologies in

¹We use 'Deaf' refers to individuals who identify as part of the Deaf community and culture, and 'deaf' refers to physiology hearing loss.

such families [6, 33, 34]. The use, perceptions, and expectations of communication technologies in mixed-hearing families are still under-investigated.

Overall, we aim to seek the answers to the research questions: What communication dynamics arise in mixed-hearing families between parents and children? How do mixed-hearing families use technologies and what do they expect from these technologies for communication?

We conducted semi-structured interviews with 11 parents from mixed-hearing families, including 5 hearing parents with DHH children (denoted as H_p) and 6 DHH parents with hearing children (denoted as D_p). Our findings highlight communication in mixed-hearing families mainly relies on spoken language, often overlooking sign language. H_p families emphasize the spoken language education, which contribute to the low usage of AAC technologies, alongside expectations for technologies for spoken language education. Meanwhile, D_p families are more willing to use AAC technologies in communication, but distrust sound awareness systems.

Our study provides empirical insights of (1) understanding the communication dynamics in mixed-hearing families and (2) D_p 's and H_p 's perceptions of current technologies used for communication, and their expectations of future technology design. (3) design insights for the future design of communication technologies.

2 RELATED WORK

2.1 Communication Technology of DHH individuals

D/deaf and hard-of-hearing (DHH) individuals, identify as physiologically deaf, culturally Deaf, or both [29, 61, 71], mostly share a Deaf culture centered on sign language [26, 31, 66, 73]. However, since most of communication in the hearing world is auditory, hearing loss reduces the effectiveness of communication of people with DHH [17, 76, 78], affecting their ability to understand information [5, 9, 27, 32] and deliver information [28, 60]. Thus, when DHH individuals communicate with hearing people via spoken language, they face unique dynamics. [50, 63, 70, 84] Many technologies are developed and employed to enhance the communication between DHH and hearing individuals, including Assistive Listening Devices (ALDs) [64, 65, 82], Augmentative and Alternative Communication (AAC) tools [2, 3, 42–44], Sound Awareness Systems [38, 39], and so on. Among these, ALDs are primarily used but fail to fully restore perception, especially in demanding listening situations such as noisy environments [5, 11, 57] and competing speakers [58], where DHH individuals use other technologies to assist.

2.2 Communication Technology of Mixed Hearing Families

Mixed-hearing families refer to families in which the hearing statuses of family members differ [63]. These include families with hearing parents and DHH children and that with DHH parents and hearing children.

Over 90% of DHH children are born into hearing families, where parents have limited knowledge of hearing loss and sign language [32]. Many families rely on spoken-only communication to integrate

their child into the hearing world [50]. However, recent studies have indicated that such communication mode is an "extremely risky proposition" in H_p families [25, 26]. DHH children in such environments are often exposed solely to auditory stimuli after birth, increasing this risk [26]. Therefore, creating sign language exposure remain to be a big challenge for parents as most hearing parents can not sign [6, 25].

Similarly, over 90% of DHH parents raising hearing children [59], there is scarce study addressing the situation of their communication. Communication challenges and issues related to spoken language are found by previous study in these families. For instance, Children of Deaf Adults (CODAs) face challenges caused by the proficiency of their DHH parents' spoke language because these parents choose to use merely spoken language in their communication [20, 46]. There are also concerns about hearing children's spoken language development in sign-only families but research showed that with more than 5–10 hours of spoken language exposure per week, children can develop spoken language similarly to their peers [16]. In addition to language related challenges, communication in these families resembles cross-cultural families, therefore, cross-cultural communication issues between hearing children and parents with DHH. [70] These issues can lead to misunderstandings and oppression [30, 46], like raising "foreign" children [70].

Moreover, studies found that in these families, parents with DHH are more likely to lack a sense of parenting competence [46, 70] due to limited access to information [40, 46, 49, 63, 70] and their own childhood experiences [70], and thus resulting in worse parenting and more conflicts [63].

While studies have investigated the use and demands of communication technologies in various mixed-hearing context settings, such as video conferencing captions [56], as well as co-located communication and collaboration [53, 80], only few studies focusing on mixed-hearing family settings. Studies have explored mixed-hearing-family-based communication augmentation tools, such as face-to-face sign language communication augmentation between hearing parents and children with DHH [33, 34] and tools informing DHH children's focal point [6]. However, these studies restrict their scope to the augmentation of sign language communication, and is between hearing parents and DHH children. Despite these, the existing communication dynamics, challenges, as well as technology practices and expectations still remain to be revealed by research, especially for these families with DHH parents and hearing children. Therefore, this paper strives understand the nuances in above two types of mixed-hearing families' communication dynamics, with or without technology assistance. Based on that, we can inform their current use and expectation of technology and discover design space and technology opportunity for facilitate the communication in these families.

3 METHODS

3.1 Semi-structured Interview

We conducted semi-structured interviews with six DHH parents raising hearing children ($D_1 - D_6$) and five hearing parents raising DHH children ($H_1 - H_5$). Table 1 provides a summary of participants' demographic and biographical information.

Participants were recruited using convenience sampling and snowball methods [13] on social media platforms, such as WeChat² and RedNote³. Each participant received a compensation of 100 CNY (approximately 14 USD). All six DHH participants were interviewed via chat message conversations for their convenience and the five hearing participants were interviewed through video conferencing tools (Tencent Meeting). All video interviews are recorded and transcribed through Tencent Meeting. The average duration for both groups was approximately 65 minutes. The transcriptions are manually cleaned, and any identifiable information is removed before data analysis. Before formal interviews, we created a preliminary version of the interview guide and conducted pilot studies with 2 participants. Pilot participants are recruited using the same method as the formal study. Based on their feedback, we clarified some questions and supplemented the questions in the section on technology usage. The study proposal and protocol passed the review of the Institutional Review Board of the author's institution prior to any research activity.

The interview questions include mixed-hearing families' language practices, and communication dynamics, such as communication modes, frequency, and strategies in their daily interaction. We also collect their communication challenges and technology practices during daily communication, including their use, perceptions, and expectations of specific technologies to facilitate communication. To grasp their technology perceptions, we employed the technology probe method [36], showing participants images and descriptions of the tools listed in Table 2 to gather their perceptions about these existing technologies.

3.2 Data Analysis

Our research team included members with extensive accessibility experience and a DHH researcher with cochlear implants (CI), encouraging a more inclusive and less biased approach to data analysis. The researchers implemented two cycle thematic analysis [12] during data analysis. In the first cycle of coding, researchers coded the data separately, using open coding [74] method and achieved an agreement after reviewing the initial codes together. In the second cycle of coding, the two researchers collaboratively group the codes found in the first cycle into themes. After 2 cycles of review and revision, a total of 257 codes and 297 codes in 3 themes were found for D_p and H_p families respectively.

4 RESULTS

Based on our research questions, we identified three themes from the interviews: (1) communication dynamics in both families, (2) technologies use, perceptions, and expectations of technologies in H_p families and (3) technologies use, perceptions, and expectations of technologies in D_p families.

4.1 Parents and Children's Communication Relies on Spoken Language

Communication modes in mixed-hearing families could be categorized as spoken-only, sign-only, and spoken-plus-sign according to previous studies [28, 48].

All families primarily used spoken language in families, despite DHH individuals in H_p and D_p families having different proficiency levels in spoken language (2 DHH parents reported poor spoken language skills, while one DHH parent and a child couldn't speak). Most families used spoken-only ($N=8$), while the other 3 D_p families communicated through spoken-plus-sign. Even in those families, spoken language was used far more frequently than sign language, because using sign language was less convenient for requiring both hands ($N=2$). D1 was concerned that his poor spoken language would negatively affect his child, mentioned *"I use spoken more than sign language to communicate with my child. My poor spoken language negatively affect my children's spoken language development, but there's nothing I can do about it."*

H_p and D_p all reported dedicating significant time and effort to spoken language education to help their children acquire spoken language as their first language. Especially for DHH children, after they attended mainstream hearing schools, H_p replaced therapists and provided spoken language education to DHH children.

In contrast to spoken language, both H_p and D_p families tended to neglect sign language education. In H_p families, only H5 has attended training courses to learn sign language but found it difficult for her, *"I really can't remember (sign language words and grammar). I can only sign some self-created gestures [...]. For example, when I let my child get a bowl, I pose with my hands like a bowl, which is a very simple word. The more complex sentences he wouldn't understand."* She never signed with her child, due to concerns that using incorrect signs might confuse her child. As her DHH child only had limited proficiency in spoken language, unfortunately, their communication was limited to basic living words. In total, communication in D_p and H_p families mainly relied on spoken language, contributing to the limited depth of communication.

4.2 Technology Use, Perceptions and Expectations in Families of Hearing Parents and DHH Children

H_p avoided AAC technologies, due to concerns about hindering children's spoken language development. When asked about AAC technologies (in Appendix Table 2), H_p expressed a positive attitude towards captions (T6 in Table 2) ($N=2$). However, they considered captions not suitable for children currently because their children were either illiterate or had low literacy. They argued that presenting captions symbolically or graphically could help DHH children. However, the other 3 H_p avoided using AAC technologies, viewing them negatively as *special strategies* to be avoided. Although H_p considered assistive expression tools (such as T2 in Table 2) to be beneficial for DHH children to communicate with parents, they worried that these tools would reduce the children's motivation to use spoken language to express themselves, potentially harmful to DHH children's spoken language development in the long term. Moreover, some H_p worried that using AAC technologies might hinder their children's integration into mainstream

²WeChat is a messaging and social media platform extensively used by people for daily communication in China

³RedNote is an image-based social media platform widely used in China

Table 1: Demographic Information of DHH and Hearing Parent Participants. Note: D1-D6 represent DHH Parent Participants, and all parent participants with DHH are self-identified as Deaf; H1-H5 represent Hearing Parent Participants.

ID	Gender	Age	Hearing Status	Education Level	Partner's Hearing Status	Children's Hearing Status	Children's Age(s)
D1	M	39	DHH Level I	Master	DHH Level II	Hearing	2
D2	F	35	DHH Level I	High School	Hearing	Hearing	2, 5, 9
D3	F	44	DHH Level I	Master	DHH Level I	Hearing	3
D4	F	33	DHH Level I	Bachelor	DHH Level I	Hearing	1, 6
D5	F	29	DHH Level I	High School	DHH Level I	Hearing	2
D6	F	36	DHH Level III	Associate	Hearing	Hearing	3
H1	M	28	Hearing	Bachelor	Hearing	DHH Level I	3
H2	F	32	Hearing	Master	Hearing	DHH Level I	3, 3 (hearing)
H3	F	31	Hearing	Middle School	Hearing	DHH Level I	3, 8 (hearing)
H4	F	29	Hearing	Bachelor	Hearing	DHH Level II	5
H5	F	28	Hearing	Elementary School	Hearing	DHH Level I	9

Table 2: Technology Probes

ID	Tools	Function	Description
T1	StorySign [1]	SL translation & animation	Translates picture books and generates sign language (SL) avatars for Deaf and Hard of Hearing (DHH) children.
T2	iConverse [83]	Assist in self-expression	Uses customized demand cards to help children express daily needs.
T3	Sound Match [23]	Sound sensitivity training	Lets users match two identical sounds by flipping sound cards.
T4	Rule the School [21]	Self-advocacy assistance	Encourages children's self-advocacy by assessing and practicing knowledge about hearing loss and amplification systems.
T5	Starkey [72]	Hearing loss simulation	Helps parents understand their children's hearing loss through situation-based simulation.
T6	Voibook [79]	Captions & translation	Provides face-to-face automatic language recognition, translation, and language training modules for DHH users.
T7	Haimababi [24]	Children status monitoring	Uses sound awareness technology to help DHH parents monitor their children's status.

hearing society (N=2). H3 stated, "(In communication) we don't use any special strategies, because when he goes out (to mainstream hearing society), others won't give him these special treatments." Besides, H_p believed spoken language education was successful, eliminating the need for AAC technologies (N=3). They claimed that they now treated DHH children as hearing in spoken language communication, as children had attended or finished spoken language education. H3 mentioned, "I now treat him as a hearing child, who can communicate with me in exactly the same way as hearing children." However, H_p reported some communication barriers in their daily life, which conflicted with the claims. H_p have placed more emphasis on their children's spoken language development than communication abilities within the family.

H_p expressed high expectations for spoken language education technologies. H_p expected assessment tools to better assess and understand their children's rehabilitation stages and

spoken language development at home, which can help them personalize spoken language education. H4 mentioned, "We rely mainly on teachers to test her (the DHH child) with rehabilitation stages and language development, but this usually happens only at the end of the semester. As a result, we know very little about what stage she's currently at."

H_p expressed the need for a systematic and professional system that provides knowledge on language education according to children's stages of language development. H4 mentioned, "I would like to have a more comprehensive knowledge system from which I can find knowledge matching my child's stage, so I can give her some proper practice."

Besides, H_p mentioned that audio-based education technologies could be better tailored to DHH children's hearing conditions. H2 pointed out differences in DHH children's ability to distinguish sounds at different frequencies. Tailored audio-based educational technologies can play audio at personalized frequencies, making it

easier for DHH children to understand and improving engagement and teaching outcomes.

4.3 Technology Use, Perceptions and Expectations in Families of DHH Parents and Hearing Children

D_p need technologies to assist spoken language education. D_p reported high usage of audio-based technologies in spoken language education. Notably, they all used pen readers to assist in the spoken language education of their hearing children. Pen readers were effective in helping children learn Mandarin pronunciation and words. D4 mentioned, *"Because we can't read and speech fluently to the children, the pen reader is quite useful to us."* However, D_p complained that the pen readers were only compatible with specific books, requiring them to purchase multiple pen readers to access more education content. So they expected pen readers to be used with unlimited books.

D_p found their children interested in talking with AI audio agent products at home ($N=2$), such as the Baidu Language Assistant [7] and Tmall Genie [4], which were not currently used for educational purposes. D_p believed that AI audio agents could assist in spoken language education, offering a wider range of knowledge through spoken language.

D_p used AAC technologies to facilitate communication with children. Beyond spoken language education, DHH parents used caption applications with their children in daily communication. They frequently used caption applications, such as Voibook [79] (T6 in Table2). Caption applications were used to transcribe hearing children's speech, especially when children talked on unfamiliar topics and lengthy sentences. Notably, the DHH father in D4's family couldn't speak and his son could only sign simple words. His communication with his child heavily relied on captions and Text-to-Speech applications on his phone.

D_p reported several drawbacks of captions. D_p noted that current captions applications performed poorly in dialect, noise, or multi-speaker scenarios, where accurate captions are most needed ($N=2$). Additionally, D1 noted that caption applications couldn't recognize young children's speech, likely due to children's immature pronunciation characteristics. Since caption applications were installed on mobile phones, D_p had to constantly check their screens, reducing eye contact with their children when communicating with their children. D3 noted, *"It feels awkward to keep looking at your phone during a face-to-face conversation."*

D_p didn't trust sound awareness systems due to inadequate alert. Besides captions, D_p still sought help from other technologies in limited visibility scenarios, especially sleeping. In the sleeping scenario, D_p frequently used sound awareness technology, such as Haima Babi (T7 in Table2), to monitor their children's physical status and nighttime crying ($N=4$). After detecting a crying or another emergency in the night, the sound awareness system would send a haptic alert to wake parents up, usually through wearable devices like smartwatches and wristbands. Participants expressed distrust in this alert method ($N=3$). D1 commented, *"Sometimes I sleep so deeply that I'm so afraid the wristband's vibration won't wake me."* Besides, D2 mentioned, *"You have to wear (wristbands) very tight to feel the vibrations, otherwise you can't get it."* Thus, even with

the sound awareness system, most of them still slept with ALDs to ensure hearing their child's cries or needs during the night ($N=3$). For instance, D3 continued using ALDs while sleeping, even though her child was already 8 years old. Sleeping with ALDs was absolutely uncomfortable and made it difficult for them to fall asleep. D2 talked, *"Sleeping with ALDs is inconvenient and uncomfortable and ALDs still needs to be charged. I've endured for years, hoping that ALDs would eventually enable convenient and comfortable 24-hour wear."* Sleeping with ALDs indicated that D_p didn't trust sound awareness technologies.

Additionally, D1 described another specific situation where visual modality was restricted: driving (the young child must be seated in a rear-facing car seat). Since he used lip-reading to comprehend speech, it became impossible for him to communicate with the child while driving, which really frustrated D1, as he sometimes need to drive alone with his child.

5 DISCUSSION

5.1 Identity Plays a Role in The Use of Technology and Depth of Communication in Mixed-hearing Families

Our study indicated that D_p who self-identified as Deaf were more willing to use technologies to facilitate communication compared to H_p families. However, the neglect of sign language education in D_p families reflects their belief that their children should belong to a different community, which in turn suggests a weak identification of their children as Deaf. Similar to Singleton et al. [70], the weak identification of children pushed D_p to rely on limited spoken language while neglecting proficient sign language to communicate with their children, resulting in restricted and asymmetrical communication. In this case, AAC technologies played an important role in these family communications. For example, in the D4 family, the father could only communicate with his child via the captions application. Studies have argued that captions cannot fully convey non-verbal aspects of speech, such as emotion, prosody, and tone [18, 19, 68], which probably lead to communication difficulties and misunderstandings in families.

In H_p families, the weak identification of DHH children is even more pronounced. Similar to Hall et al. and Lederberg et al. [25, 50], hearing parents invested a significant amount of time and effort in spoken language education, with the hope that DHH children would "overcome" hearing loss and integrate into hearing society [50]. Despite using ALDs, DHH children still experience biological hearing differences from hearing children, and these differences cannot be fully eliminated [50]. This creates a mismatch between parents' cognitive identity of children and children's biological identity, potentially pushing parents to intentionally overlook the barriers and demands of technology in communication, further hindering effective communication in the families. Besides, as AAC technologies offer communication alternatives to DHH children, reducing the motivation for children to practice hearing and speaking. Possibly as a result of this, hearing parents avoid using AAC technologies in family communication.

5.2 Design Insights and Opportunities for Assistive Communication Technologies

Communication Technologies Should Consider Low Literacy of Children. From Section 4.3, we found that DHH children could not use captions in communication due to their limited literacy. Previous research [15] indicated that even a single typo could affect the comprehension and memory of the sentence for DHH individuals with low literacy and cognitive abilities. Therefore, when designing technologies for DHH children, it is crucial to consider literacy rates and incorporate multiple design features, such as symbols, emojis, or other visuals to enhance captions.

Sound Awareness Need More Clear And Effective Alerts. We suggest that sound awareness technologies designed for mixed-hearing families should feature clear and effective alerts. Similar to [39], which mentioned the need for alerts from DHH individuals in the sound awareness systems, our findings in Section 4.3 show that insufficient alerts can significantly reduce parents' trust in sound awareness technologies. Hearing children, who have not yet developed basic living skills, struggle to gain the attention of their DHH parents. To some extent, the alerts are emergency calls to the parents in place of the children. Haptic alert through wearable devices is an usual method in sound awareness system [24], which is not robust according to our participants. As many studies about sound awareness focus on how to discern and represent sound [38, 39], We suggest alert also should be taken seriously to design the robust and scientific alert system. For example, if the haptic alert don't work at begin, try a flashlight to wake DHH parents up. When designing a sound awareness system, the effectiveness of the alert system in the different scenarios should be carefully assessed.

Sign Language Education Technologies for Hearing Parents. Previous studies have emphasized the importance of sign language and encouraged sign language education in H_p families [25, 26]. Our study revealed the overlook of sign language, suggesting that the asymmetry in proficiency levels between hearing parents and DHH children may contribute to its limited use. Learning sign language is difficult for hearing parents. As young DHH children also in the process of learning sign language, we propose that sign language education tools can encourage hearing parents to learn sign language alongside their DHH children, either as co-learners (both parent and child as classmates) or through a model where DHH children take on the role of teacher. For example, a mini-game where using sign language competes for rewards.

Spoken Language Education Technologies Enable Involvement of DHH Parents. According to our findings, D_p used pen readers and AI agent products to assist spoken language education to their hearing children. The two technologies were based on voice interaction, requiring D_p to carefully listen to join the interaction or give help to children. Interactions between parents and young children have been proven to be beneficial for children's language learning [77]. We believe that technologies can be designed that provide spoken language to children through audio, while also incorporating visual elements, captions, TTS, or other components to ensure DHH parents' involvement in the educational process.

5.3 Cultural Perspectives' Influence on Parents' Perception of Technologies in Mixed-Hearing Families

All of our participants were from China. Previous studies have indicated parents in China prioritize spoken language education for their children, beyond sign language and communication in mixed-hearing families [14, 20]. Their emphasis on spoken language education leads them to favor technologies that support the spoken language development of children. Moreover, due to the relatively low adoption rate of assistive technologies in China, parents have limited experience or misperceptions with assistive technologies [51, 62].

5.4 Limitations

Our study has several limitations. First, we only interviewed 11 Chinese parents from mixed-hearing families, and this small sample size cannot fully represent the broader population of mixed-hearing families. Furthermore, our participants generally were highly educated on average (as shown in Appendix Table 1), which may not reflect the broader mixed-hearing family population, as high educational attainment is less common in both hearing and DHH populations [75]. Lastly, we didn't reach the families where hearing parents with DHH children communicate via sign language, thus we lack insights into sign language education or communication within these families—an important aspect of mixed-hearing family dynamics.

6 CONCLUSION

Our qualitative study explored communication dynamics and technology use in mixed-hearing families. We found that two types of mixed-hearing families had common in communication pattern, where spoken language dominated, with sign language often overlooked. But they showed absolutely different attitudes and expectation of technologies. For example, hearing parents in H_p families resisted using AAC technologies because they fear hindering children's language development, while parents of DHH frequently used them for family communication. We discussed the weak identification of children as "Deaf" was responsible to lower adoption of assistive technologies and limited communication depth. According to our findings, we offered design insights for AAC, sound awareness, and sign language education technologies. This study shed lights on communication dynamics and highlighted demands for more studies in mixed-hearing families.

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A INTERVIEW GUIDE

A.1 Basic Family Information

A.1.1 Personal Information. First, we would like to understand more about your background and your child's basic information.

- Your gender, age, education level, and occupation.
- (For D_p) What level of hearing loss do you have (dB)? When was it diagnosed, and when did you begin intervention?
- (For D_p) Do you use assistive listening devices like cochlear implants or hearing aids?
- (For D_p) How proficient are you in spoken language and sign language? Do you know Chinese Sign Language (CSL)?
- (For H_p) How proficient are you in sign language?
- How much time do you usually spend with your child each day?

A.1.2 Child's Information. Please provide details about your child.

- Child's gender, age, and educational level.
- How proficient is your child in spoken language and sign language?
- (For H_p) What level of hearing loss does your child have (dB)? When was it diagnosed, and when did you start interventions?
- (For H_p) Is your child using any assistive listening devices (e.g., hearing aids, cochlear implants)? How is their current hearing condition?
- (For H_p) Is your child attending a regular school or a special education school? What grade is your child in?
- (For D_p) Does your child have hearing siblings? If so, please provide their gender and age.

A.2 Communication

A.2.1 Communication Methods.

- How do you generally communicate with your child—using sign language, spoken language, or a mix of both?
- Could you describe in detail how you and your child usually communicate on a day-to-day basis?
- Do you use any assistive tools or technology for communication? What purpose do they serve?
- If you use a mix of communication methods, When do you prefer to use sign language, and when do you prefer spoken language?

A.2.2 Communication Challenges.

- Have you experienced communication asymmetry (e.g., you and your child using different languages)? How do these situations affect interaction or emotional connection—do they create barriers or enhance communication efficiency?
- How did your child learn sign language and/or spoken language? What tools or techniques were helpful?
- Has your communication method with your child changed as they have grown? For example, are you more inclined to use spoken language at certain stages?

- At which stages did the communication method change?
- How have the frequency and content of communication with your child changed as they have matured?
- After your child reached the age of 2 and developed more awareness of communication, did you encounter any challenges or difficulties in communication compared to non-DHH families? How did you handle these?
- What tools or methods do you believe could help improve your communication challenges with your child?

A.3 Family Daily Activities

A.3.1 Types of Activities. Daily family activities refer to regular, routine interactions such as putting your child to sleep, eating meals, or going to the bathroom.

- Could you describe your interactions with your child during these daily activities (e.g., doing chores, shopping, cooking)?
- How do you communicate with your child during these activities—using sign language, spoken language, and do you use any tools or technologies?
- In what ways do you help your child understand and complete tasks during these activities?

A.3.2 Communication Technology in Daily Activities.

- Do you use any specific communication technologies or tools during these activities? What are their types and functions (e.g., sound awareness tools)?
- How do these tools aid daily activities, and are there any ways they create obstacles?

A.3.3 Challenges and Solutions.

- Have you encountered any challenges or difficulties during these daily tasks? How did you resolve them? Did you use any specific tools or technologies?
- What tools or strategies do you believe could better help you manage these challenges?
- Could you describe a particularly successful example of co-operation during daily activities? What activity was it, and why do you consider it successful?

A.3.4 Special Cases of Cooperation.

- Are there specific tasks or situations where cooperation is especially important, such as crossing the road safely or solving a complex problem? How do you and your child divide tasks and communicate during these situations?
- In summary, how do you feel your (or your child's) hearing impairment has impacted your interactions and cooperation during daily activities? Do you believe your current strategies are well-suited to your child?

A.4 Parent-Child Activities

A.4.1 Types of Activities.

- How much time do you typically spend each day on parent-child activities?
- What types of activities do you and your child engage in (e.g., reading picture books, playing with building blocks)? Do you use spoken language, sign language, or a mix during these activities?

- Do you use any communication technologies or tools during these activities? What are they used for?

A.4.2 *Changes Over Time.*

- How have these activities changed over time as your child has grown (e.g., more time spent on language education activities)?
- Have you faced any challenges during these activities? How did you overcome them? Did you use any tools or technologies?

A.4.3 *Successful Examples of Cooperation.*

- Can you describe a particularly successful parent-child activity? What was the activity, and why do you think it was successful? How did you and your child cooperate during the activity?
- Are there specific activities that require significant cooperation or interaction, such as solving a problem or completing a project together? How did you and your child collaborate on this task?
- How do you assess your overall cooperation with your child in these activities? How has your (or your child's) hearing impairment impacted this cooperation?

A.5 Summary and Outlook

- Despite the tools or technologies mentioned earlier, are there any additional tools or strategies you use?
- Are you satisfied with the current communication and interaction between you and your child? Why or why not? How would you like to improve it?
- What other tools or technologies do you think could help improve communication and interaction with your child?

A.6 Follow-up Questions

A.6.1 *Hearing and Communication Preferences.*

- Do you mainly use spoken language with your child? Why do you prefer spoken language?
- (for D_p) Have you noticed any speech issues in your hearing child?
- How did you discover these issues, and what measures have you taken?

A.6.2 *Child's Understanding and Adaptation.*

- Have you informed your child about your (or their) hearing loss?
- Does your child understand your (or their) hearing condition? If so, how does this affect their communication (e.g., speaking slower or louder)?
- (for D_p) Do you want your child to adapt their communication style for you?

A.6.3 *Communication Tools and Strategies.*

- Have you used any tools or technologies to help with communication (e.g., reading pens)?
- In noisy environments, how do you manage communication with your child?

A.6.4 *Language Development.*

- How do you teach your child language (e.g., spoken or sign language)?
- How do you compare your child's learning speed between spoken language and sign language?

A.6.5 *Communication Depth.*

- Do you ever engage in deep, meaningful communication with your child (e.g., heart-to-heart conversations)? If so, do you prefer spoken or sign language?